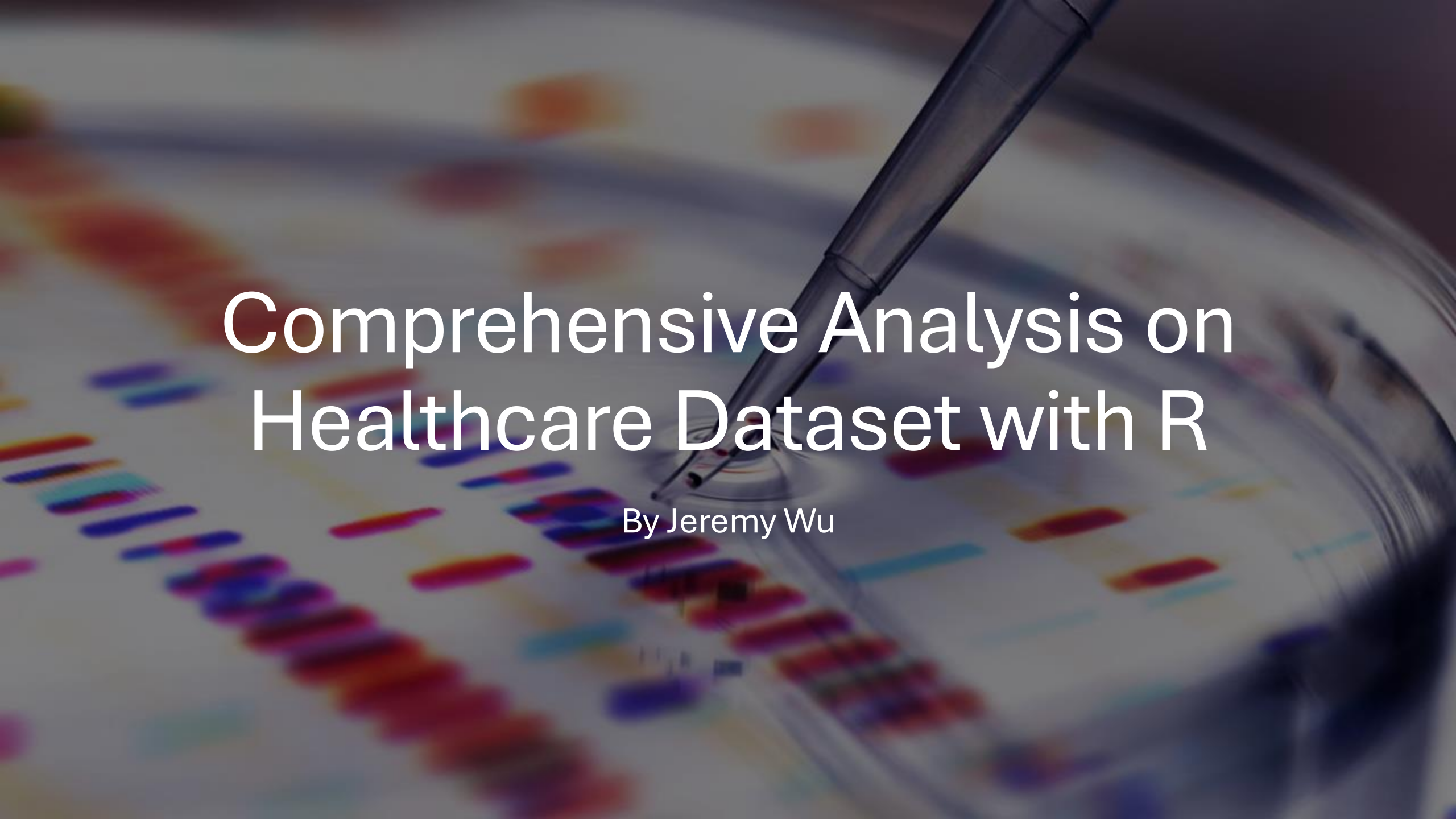


The background of the slide is a blurred image of a DNA microarray. It shows a grid of small, colorful spots (red, blue, green, yellow) on a white surface. A glass pipette tip is visible in the upper right, positioned over one of the spots. The overall image has a soft, out-of-focus quality.

Comprehensive Analysis on Healthcare Dataset with R

By Jeremy Wu

About the dataset

- 55,500 number of rows, each row represents an individual patient
- 15 columns:
 - Patient's name
 - Age
 - Gende
 - Blood Type
 - Medical Condition
 - Admission Date
 - Doctor acquainted
 - Hospital Name
 - Insurance Coverage for each patient
 - Total billed amount
 - Room Number
 - Admission type
 - Discharged date
 - Medication provided
 - Test Result

Research Questions

Testing relationships

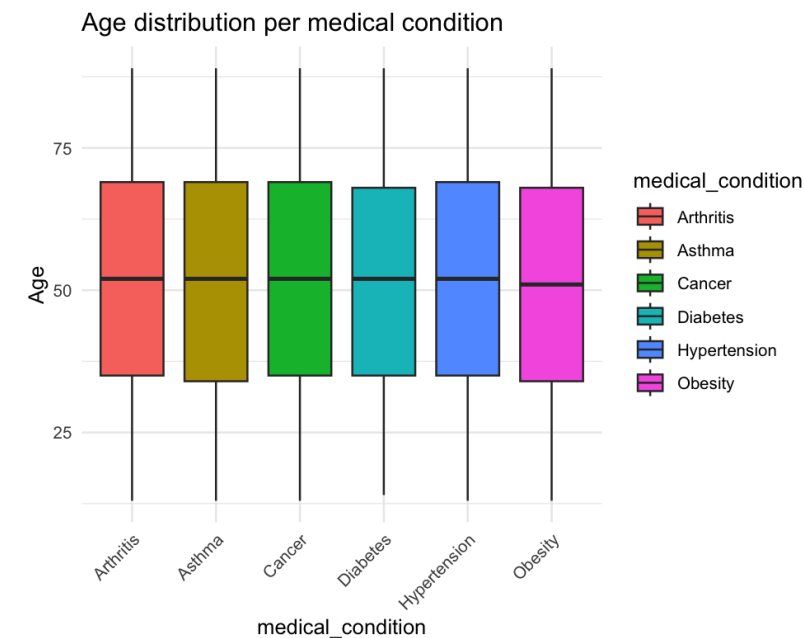
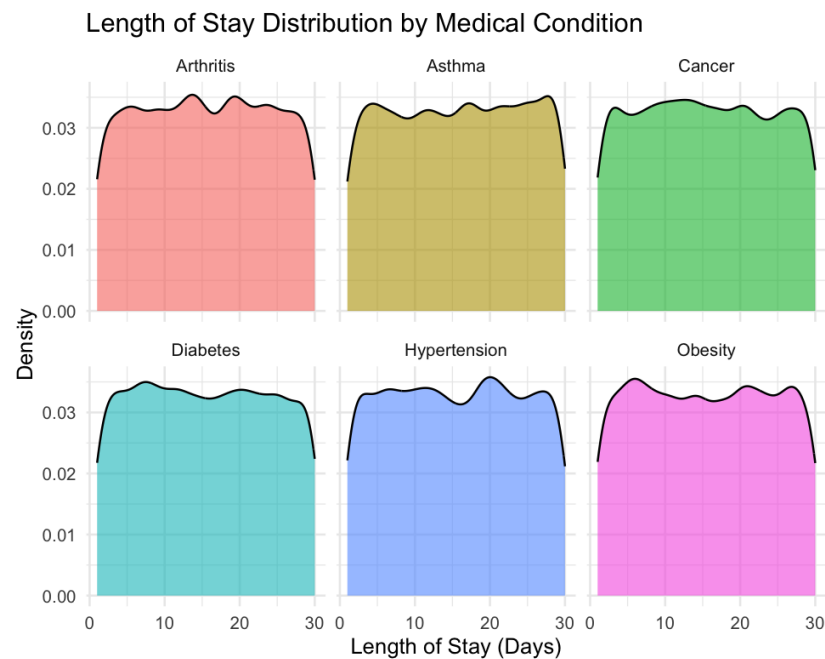
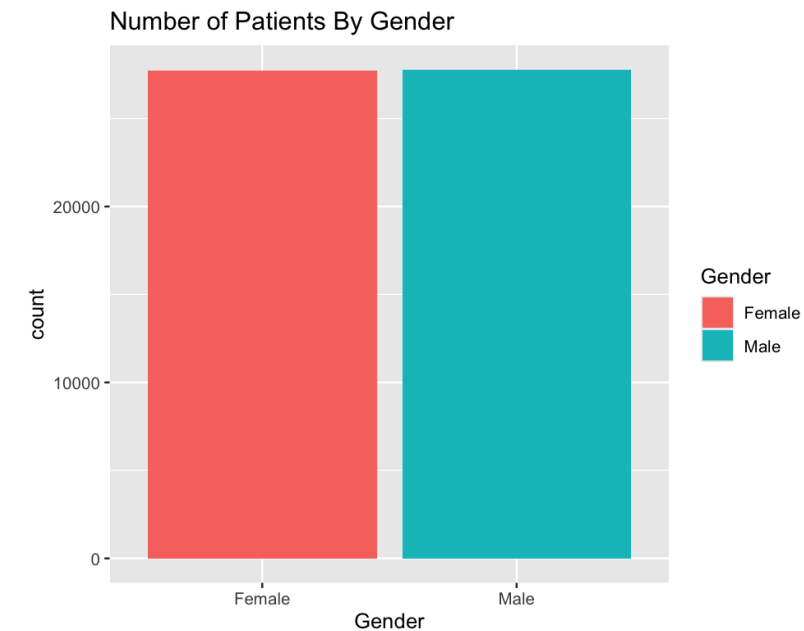
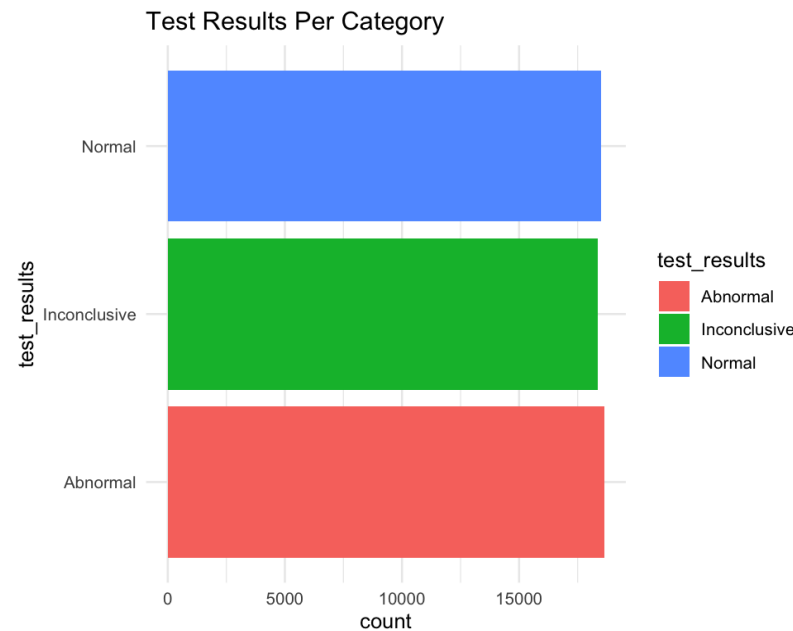
Main research question: What factors affect billing amount the most, and what steps can we take to minimize this amount for patients

- Does admission type, medical condition, age, test result or length of stay affect the billing amount? If so, which independent variable has the most significant impact in determining the final billing amount?

Other research questions:

- What are the test result proportions from our dataset?
- What is the age distribution per medical condition?
- Does length of stay at the hospital vary by medical condition?
- Does the insurance provider affect the average length of stay for each patient?
- Does gender, age, blood type, or medical condition significantly affect the test result with a baseline of "normal"?
- Does the type of admission predict the length of stay in the hospital?

Part 1: Visualizing our dataset



Hypothesis Testing

1. One sample t test on age & billing amount at a 95% confidence interval

Age Hypothesis --> $H_0: \mu = 51.5$ | $H_1: \mu \neq 51.5$

Results --> p value (0.64) > α (0.05) = failed to reject H_0

Billing Amount Hypothesis --> $H_0: \mu = 25539$ | $H_1: \mu \neq 25539$

Results --> p value (0.99) > α (0.05) = failed to reject H_0

2. ANOVA test on whether each medication type affects the total billing amount ($\alpha = 0.05$)

H_0 : Mean billing amount is the same across all 5 medication types

H_1 : At least one medication type has a significantly different mean billing amount

F value = 1.139 | p value (0.336) > α (0.05) = failed to reject H_0

Does insurance providers affect the length of stay for patients?

Model: Linear Regression ($\alpha = 0.05$)

H_0 : Insurance provider does not significantly predict length of stay

H_1 : At least one insurance provider significantly predicts length of stay

	Estimate	Std. Error	t-value	p-value
(Intercept)	15.4474	0.0829	186.350	< 2e-16
Blue Cross	0.0747	0.1168	0.639	0.523
Cigna	0.0385	0.1164	0.331	0.741
Medicare	0.1856	0.1166	1.592	0.111
UnitedHealthcare	0.0079	0.1167	0.068	0.946

R^2	Adjusted R^2	F-statistic	p-value	df	Residual Std. Error
0.0000607	-0.000011	0.843	0.4976	55495	8.66

Does gender, age, blood type, or medical condition significantly impact test results?

Model: Multinomial Logistic Regression ($\alpha = 0.05$)

H_0 : Gender, age, blood type, and medical condition do not significantly predict test result outcome

H_1 : At least one predictor significantly influences whether a test result is Abnormal or Inconclusive compared to Normal

Baseline: Normal Test Result

Predictor	Coefficient	p-value
(Intercept)	0.0152	0.7477
Gender (Male)	-0.0238	0.2523
(Asthma)	-0.0980	0.0064
(Cancer)	-0.0306	0.3943
(Diabetes)	-0.0230	0.5214
(Hypertension)	-0.0880	0.0145
(Obesity)	-0.0254	0.4794
Blood (A+)	0.0147	0.7231
Blood (AB-)	-0.0115	0.7814
Blood (AB+)	-0.0025	0.9514
Blood (B-)	0.0071	0.8632
Blood (B+)	0.0121	0.7720
Blood (O-)	0.0055	0.8941
Blood (O+)	0.0158	0.7038
Age	0.0008	0.1276

Predictor	Estimate	p-value
(Intercept)	-0.0237	0.6195
(Male)	-0.0269	0.1969
(Asthma)	-0.0597	0.0980
(Cancer)	-0.0178	0.6235
(Diabetes)	-0.0309	0.3932
(Hypertension)	-0.0305	0.3963
(Obesity)	-0.0229	0.5272
(A+)	0.0315	0.4484
(AB-)	-0.0328	0.4301
(AB+)	0.0210	0.6129
(B-)	-0.0094	0.8211
(B+)	0.0553	0.1830
(O-)	-0.0047	0.9104
(O+)	-0.0046	0.9124
Age	0.0009	0.0778

Abnormal vs Normal (Baseline) Inconclusive vs Normal (Baseline)

ANOVA Test

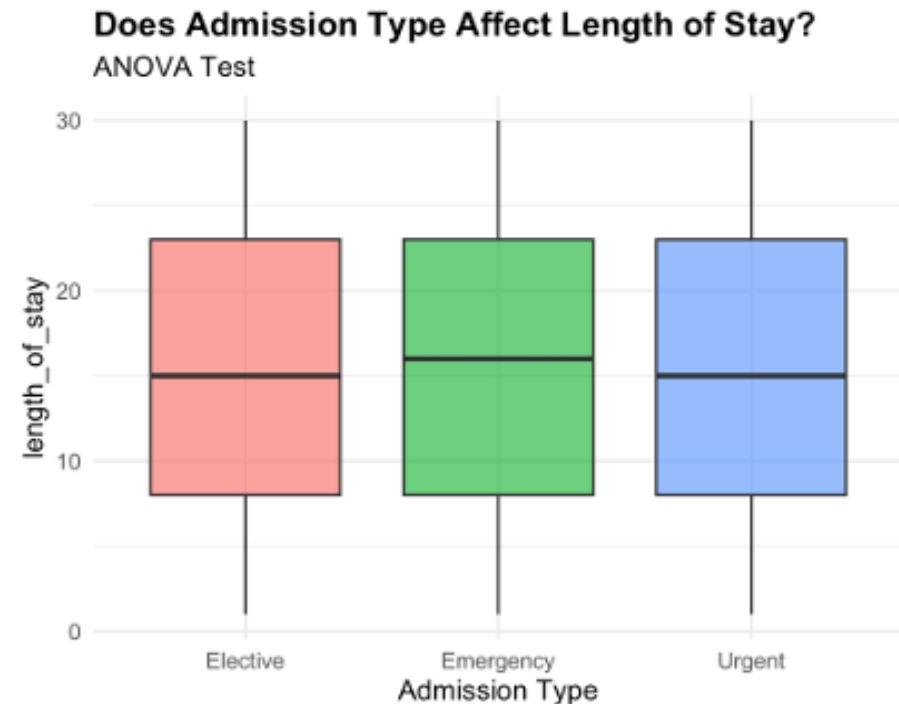
Does admission type have a significant influence on the average length of stay for patients?

H_0 : Mean length of stay is the same across all 3 admission types

H_1 : At least one admission type has a significantly different average length of stay

$\alpha = 0.05$

	Df	Sum Sq	Mean Sq	F-value	p-value
Admission Type	2	330	164.87	2.199	0.111
Residuals	55497	4161467	74.99		



Hypothesis Testing on Billing Amount

3. Linear Regression Model for Age & Billing Amount ($\alpha = 0.05$)

H_0 : Age does not significantly predict billing amount

H_1 : Age significantly predicts billing amount

R squared = 0.0000146

P value = 0.3667

Results: P value $> \alpha$ (0.05) = failed to reject H_0

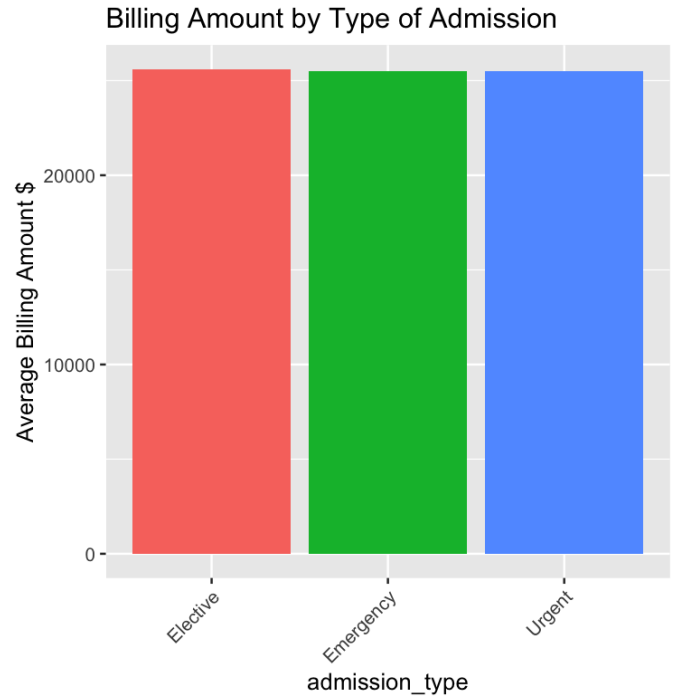
4. Two sample t test for test results & billing amount ($\alpha = 0.05$)

H_0 : The mean billing amount for Normal test results = mean billing amount for Abnormal test results

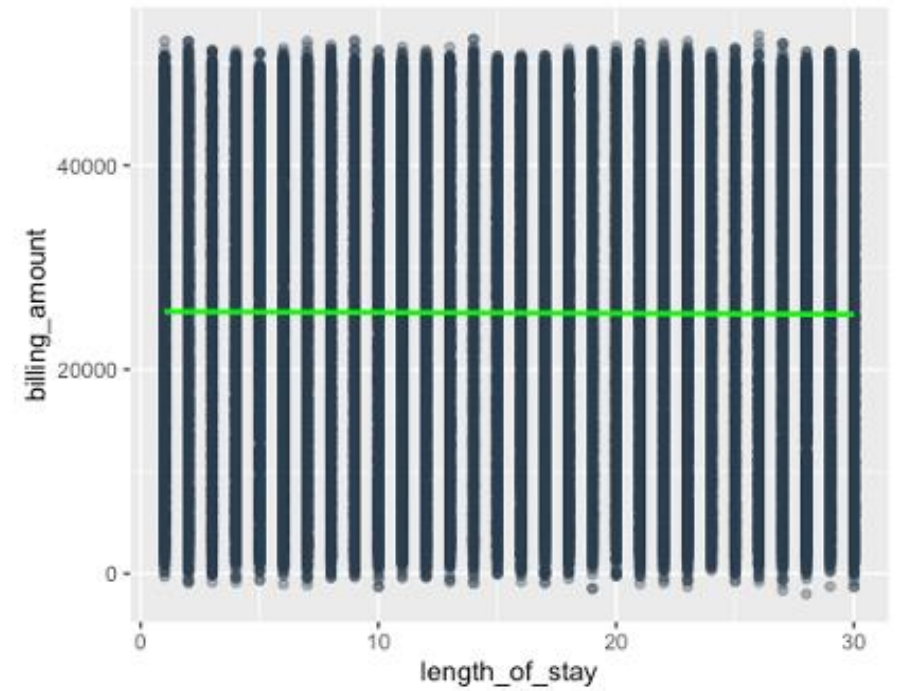
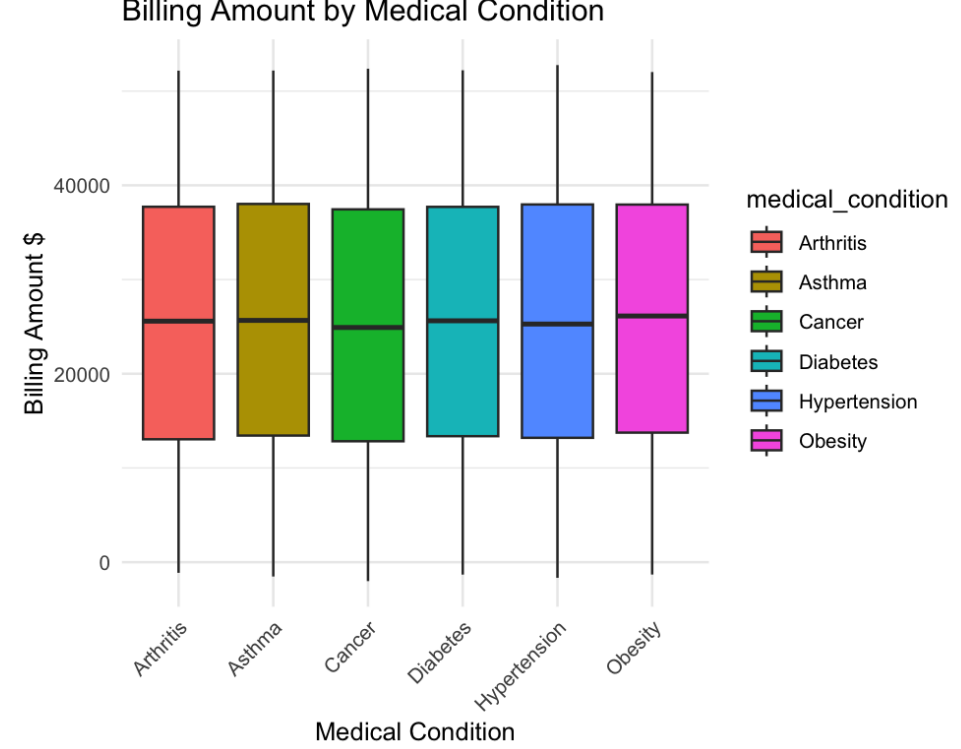
H_1 : The mean billing amount for Abnormal test results is significantly higher than Normal test results (one-tailed), or simply different (two-tailed)

Results: p value (0.58) $> \alpha$ (0.05) = failed to reject H_0

T value = 0.55



Billing Amount



Conclusion & Findings

- No variable produced a statistically significant relationship with any outcome across all hypothesis tests and regression models
 - Specifically age, medication type, admission type, insurance provider, and length of stay all failed to predict billing amount
- Dataset most likely to be synthetically generated
 - Each variable was generated independently, explaining why nothing predicts billing amount
- Real healthcare data would show meaningful relationships between these variables, specifically in predicting the outcome variable billing amount

References

Healthcare Dataset. (n.d.).

[Www.kaggle.com](https://www.kaggle.com).

<https://www.kaggle.com/datasets/prasad22/healthcare-dataset/data>

